

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Quiz-2
Duration: 20 minutes

Semester: Fall 2023
Full Marks: 15

CSE 470: Software Engineering

Name:

ID:

1. Write down code to implement a Singleton class. You can use any language of your choice.
2. What is the importance of implementing an observer design pattern?
3. Consider the following classes:

```
public interface Sandwich {  
    abstract void toppings();  
    abstract void buns();  
}  
  
class HeavySandwich {  
    public void sausage() {  
        System.out.println("Sausages added for heavy sandwich");  
    }  
  
    public void cheese() {  
        System.out.println("Cheese added for heavy sandwich");  
    }  
  
    public void bread() {  
        System.out.println("Thick crust for heavy sandwich");  
    }  
}
```

Write a class adapter for the HeavySandwich class, such that it implements the Sandwich interface. You can consider sausage and cheese as toppings, and bread as buns. Also, override the sausage function, such that the following line is printed instead: "Over Fried sausages added for sandwich."

You can use any language of your choice.